

OMKAR THAWAKAR

Abu Dhabi, UAE | +971-585-25-1761 | thawakar.omkar@gmail.com
omkarthawakar.github.io | [LinkedIn](#) | [GitHub](#) | [Google Scholar](#)

Researcher | Computer Vision | Multimodal AI | Large Language Models | Video Understanding

PROFESSIONAL SUMMARY

PhD researcher in Computer Vision and Multimodal AI with 16+ publications across CVPR, ICCV, ECCV, ICLR, ACL, EMNLP, MICCAI, and IJCV. Developed widely adopted open-source models with 1.2M+ HuggingFace downloads and 3.5K+ GitHub stars, including MobiLlama, an ICLR-SLLM 2025 Spotlight paper. Research interests include multimodal foundation models, video understanding, visual reasoning, and agentic AI systems. Holder of a US patent in video instance segmentation.

EDUCATION

Ph.D., Computer Vision

Aug 2023 – Spring 2027 | Abu Dhabi, UAE

Mohamed bin Zayed University of Artificial Intelligence (MBZUAI)

- Thesis: *Video Understanding with Large Multimodal Models* Advisor: Prof. Fahad Khan
- Research: Multimodal Learning, Vision-Language Models, Efficient LLMs, LLM Agents, Video Detection & Segmentation
- GPA: 3.74 / 4.00

B.Tech., Computer Science Engineering

Jun 2015 – Jul 2019 | Nanded, India

Shri Guru Gobind Singhji Institute of Engineering and Technology (SGGSIE&T)

- Coursework: Neural Networks, ML, Algorithms, OS, Cryptography, Distributed Systems, Computer Networking
- GPA: 8.7 / 10.0 **Best B.Tech Thesis Award**

RESEARCH & INDUSTRY EXPERIENCE

Research Scientist Intern — Multimodal Agents, Adobe Research

2026 | Ongoing

- Developing self-evolving agentic systems for GUI that combines multimodal models, tool use, and retrieval-augmented reasoning for interactive GUI experiences.
- Designing evaluation protocols for GUI agents in collaboration with Adobe Research scientists.

Researcher, Mohamed bin Zayed University of AI (MBZUAI)

May 2021 – Jul 2023 | Abu Dhabi, UAE

- Achieved SOTA on video instance segmentation with transformer-based architecture — published at **ECCV 2022**.
- Achieved SOTA on 3D mitochondria instance segmentation using transformers (+3% accuracy) — **MICCAI 2023**.
- Pioneered open-world video instance segmentation using shallow features for novel object detection — **IJCV 2023**.

Machine Learning Engineer, Chefling India Pvt Ltd

Feb 2020 – May 2021 | Bangalore, India

- Deployed ML-powered recipe scraping service using Scrapy, achieving **87% accuracy** on complex web structures.
- Built recipe tagging service covering **70+ tag categories**, eliminating external dependency and reducing costs by **92%**.

Research Assistant, Indian Institute of Technology Ropar (IIT-Ropar)

Jan 2019 – Feb 2020 | Ropar, India

- Researched image/video super-resolution, moving object segmentation, and image restoration (haze, fog, lighting).
- Developed AI solutions for educational and commercial applications; won Best B.Tech Thesis Award.

APPLIED AI LEADERSHIP

AI Technical Lead, Lawa.AI

Jul 2024 – May 2026 | Abu Dhabi, UAE

- Led the design and deployment of production-grade Agentic AI systems ([Lawa-Agent-studio](#)), and RAG AI applications for enterprises. Visit <https://lawa.app/studio> to know more.
- Developed AI products from research prototypes to scalable agentic AI pipelines, and represented UAE & MBZUAI AI innovation at ITU WTSA 2024 Innovation Xchange.

AI Systems Lead, Nutrigenics.Care

Jan 2024 – Dec 2025 | Abu Dhabi, UAE

- Developed Nutrigenics, AI platform for clinical nutrition intelligence and personalized healthcare recommendations.
- Secured competitive innovation funding (including Microsoft Founders Hub support), culminating in the platform's technology acquisition by a healthcare entity.

SELECTED PUBLICATIONS

16 total — CVPR, ICCV, ECCV, ICLR, ACL, EMNLP, MICCAI, IJCV

- **EvoLMM: Self-Evolving Large Multimodal Models with Continuous Rewards** [CVPR 2026]
- **CoVR-R: Reason-Aware Composed Video Retrieval** [CVPR 2026]
- **Thinking Beyond Labels: Vocabulary-Free Fine-Grained Recognition using Reasoning-Augmented LMMs** [CVPR 2026]
- **Beyond Simple Edits: Composed Video Retrieval with Dense Modifications** [ICCV 2025]
- **LlamaV-o1: Rethinking Step-by-Step Visual Reasoning in LLMs** [ACL 2025]
160+ citations · 70K+ HF Downloads
- **MobiLlama: Towards Accurate and Lightweight Fully Transparent GPT** [ICLR 2025]
Spotlight - Top 2%, ICLR-SLLM · 50+ citations · 352K+ HF Downloads · 600+ Github stars
- **All Languages Matter: Evaluating LMMs on Culturally Diverse 100 Languages** [CVPR 2025]
CVPR 2025 Highlight · 50+ citations · 11K+ HF Downloads
- **CAMEL-Bench: A Comprehensive Arabic LMM Benchmark** [NAACL 2025]
- **Composed Video Retrieval via Enriched Context and Discriminative Embeddings** [CVPR 2024]
- **XrayGPT: Chest radiographs summarization using large medical vision-language models** [ACL-W 2024]
340+ citations · 390K+ HF Downloads · 500+ Github stars
- **Video Instance Segmentation in an Open-World** [IJCV 2024]
- **3D Mitochondria Instance Segmentation with Spatio-Temporal Transformers** [MICCAI 2023]
- **Video Instance Segmentation via Multi-Scale Spatio-Temporal Split Attention Transformer** [ECCV 2022]
- **System and Method for Video Instance Segmentation via Multi-Scale Spatio-Temporal Split Attention Transformer, Patent #12499573, Published Dec 2025** [US Patent]

TECHNICAL SKILLS

Research Areas: Computer Vision, Multimodal Learning, Vision-Language Models, Video Understanding, LLMs, Agentic AI, Dense Retrieval, Image/Video Segmentation

Deep Learning: PyTorch, TensorFlow, HuggingFace Transformers, LoRA/QLoRA, RLHF, DPO, RAG, Diffusion Models

Programming: Python, R, Java, C

MLOps / Cloud: AWS, Azure, Google Cloud Platform, Docker, MLflow, BigQuery

Data & Web: PySpark, NumPy, Pandas, Matplotlib, Scikit-learn, Flask, Django, React-JS, MongoDB, SQLite

HONORS & AWARDS

- **Winner — National Grant Program for Culture & Creativity** — 100K AED, Ministry of Culture *Nov 2025*
- **1st Place — Khalifa Fund Innovation Competition** — 250K AED product grant, ADBW *Dec 2024*
- **ICLR-W 2025 Spotlight** — MobiLlama selected as Top 2% paper at SLLM Workshop *Feb 2025*
- **Microsoft Founders Hub Grant** — Nutrigenics.Care awarded \$150K in technical support *Jul 2024*
- **1st Place — Sandook-AI-Watan Student Pitch Competition** — 30K AED project grant *Nov 2024*
- **1st Place — Pitch Day with Microsoft** — MBZUAI Incubation & Entrepreneurship Center *May 2024*
- **Best B.Tech Thesis Award** — Video Super-Resolution using Recurrent GAN, SGGSIE&T *May 2019*
- **Best Research Paper — IC3NS-2018** — Application of ML Algorithms for IPM Device *May 2018*

ACADEMIC SERVICE

- **Reviewer:** CVPR, ICCV, ECCV, IEEE, CVF, Springer — 100+ papers reviewed from top-tier venues *Jul 2022 – Present*
- **Teaching Assistant,** MBZUAI — Department of Computer Vision *August 2023 – Dec 2025*

REFERENCES

Prof. Fahad Khan (Primary PhD Advisor)
Deputy Chair CV, MBZUAI fahad.khan@mbzuai.ac.ae

Prof. Mubarak Shah (Mentor / Collaborator)
University of Central Florida shah@crvc.ucf.edu

Prof. Timothy Baldwin (Mentor / Collaborator)
Provost, MBZUAI timothy.baldwin@mbzuai.ac.ae

Dr. Salman Khan (PhD Co-Advisor)
Assoc Prof, MBZUAI salman.khan@mbzuai.ac.ae